A Key to the Voynich Manuscript

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Abstract

The Voynich Manuscript is a 15th century text which features herbalist, pharmaceutical, balneological, and astronomical illustrations [1]. Despite numerous attempts over the past 100 years the text has so far resisted all efforts at decryption. Even the application of sophisticated statistical methods has not led to a clear identification of the language the text is written in and leaves some chance that it is perhaps an elaborate hoax [1]. Here we partially decode the Voynich alphabet and provide an identification of the Voynich language as vernacular Italian (lingua volgare) written in shorthand. We are able to show that the words we read make sense in the context of the illustrations. We provide a lexical list of 400+ words that can be consistently read throughout the text in vernacular Italian with our decipherment and show that the text displays morphological and grammatical features consistent with the language identification. While there is more work to be done in translating the text, the key to the manuscript's decipherment is now available.

Main

The Voynich Manuscript, cataloged as MS408 in Yale University's Beinecke Rare Book and Manuscript Library, has remained undeciphered since its acquisition by book dealer W. Voynich near Rome in 1910 or 1911 [1]. The manuscript is written in an unknown script and language, with no direct translation or decryption to this day despite numerous attempts including ones by specialists of the United States National Security Agency [cf. 2]. The manuscript's contents span five distinct thematic sections, including detailed illustrations of plants, astrological charts, pharmaceutical descriptions, and balneological scenes featuring human figures immersed in baths. It consists of 116 vellum folios, although several pages are missing [1, 3].

Some have proposed that the manuscript is written in a cipher [4, 5, 6] while others have suggested it possibly might be a hoax with no actual linguistic meaning [7, 8, 9]. However, several linguistic and statistical analyses have shown that the manuscript's structure is too complex and organized to be mere gibberish, hinting at a real language or system of encoding [10]. Some suggest it is written in a cipher based on a known language like Latin, Hebrew, or an early Romance language. Others, such as Bax [11] and Cheshire [12], have proposed partial decodings, claiming the manuscript uses a blend of different medieval languages, including Italian, Occitan, and Latin. Still, these attempts have remained controversial and have not been widely accepted by the academic community. A limited one line solution for Latin has been proposed by Gibbs [13] and Hauer and Kondrak [14] propose Hebrew as the language of the manuscript. Bowern and Lindemann [1] most recently point out how many of the proposed solutions exhibit "strong confirmation bias, omitting any information that does not fit the theory they propose" and focus on vocabulary while leaving out morphology and syntax [1].

Some researchers believe that the manuscript could have been created by multiple authors or scribes, which might explain inconsistencies in the handwriting and structure [15]. Radiocarbon dating of the vellum by

the University of Arizona consistently show a calibrated age in the early 15th century [16], however, the dates remain unpublished and might shift slightly with the availability of a new calibration curve [17].

The literature on the identifications of the plants shown in MS408 is extensive as these identifications might yield hints on the vocabulary appearing in the text. While some claim that the plants are not identifiable [18], others make extensive comparisons with medieval herbalist codices [19]. Perhaps most controversially, [20] identifies some of the plants to originate in the Americas. Recently, [21] have elaborated upon the idea and suggest that therefore New World languages need to be considered as well. These ideas have been challenged by [1]. The plant illustrations do appear in stylized fashion typical for 15th century manuscripts and are therefore consistent with the radiocarbon dates of the vellum.

The main problem with all previous deciphering attempts is that none of them have gone beyond individual words or lines and deciphered longer passages of text in the context of the illustrations (except for [22]). It has never been shown that a language proposition was effectively transferable to all parts of the manuscript.

In the following we will show a careful bottom-up approach in first deciphering individual words in the context of illustrations. We then develop the alphabet of MS408. Based on this alphabet, we identify words from every page of the manuscript. We identify the language and elaborate upon the grammatical and morphological aspects of the words and word combinations we read.

Results

Serendipitously, one of the first legible words which allowed us to start this deciphering attempt was the word POLAR next to a star illustration, indicating we had identified the North Star as a guiding accumulation of letters for the next steps (Figure 1a). The word consists of familiar letters which can be read: a π for P, an *o* for O, a λ for L, an *a* for A, and an *r* for R.

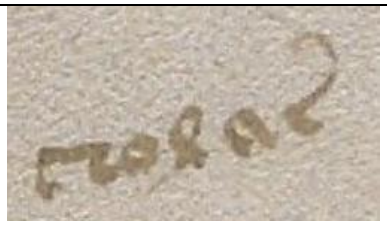
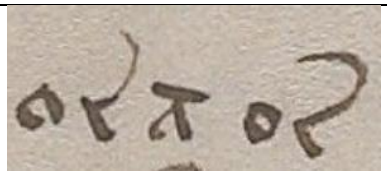
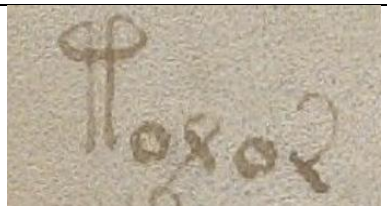
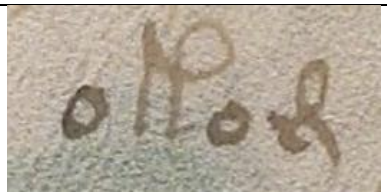
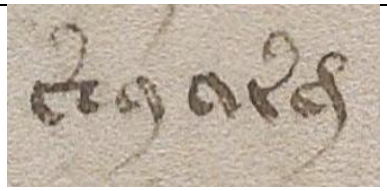
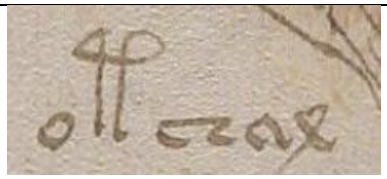
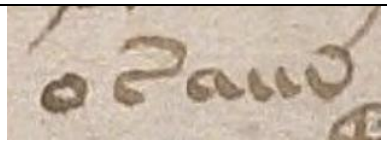
Given the initial seemingly successful attempt with identifying star names and the availability of terms for stars in the chronologically close Almagestum Claudii Ptolemei of 1528 [23], we were able to

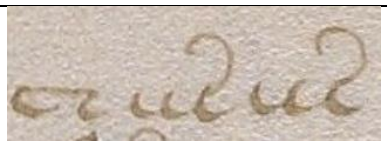
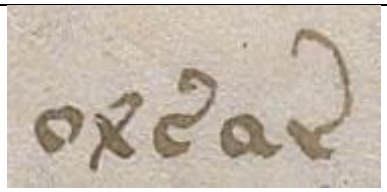
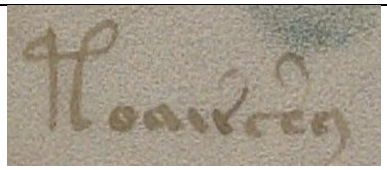
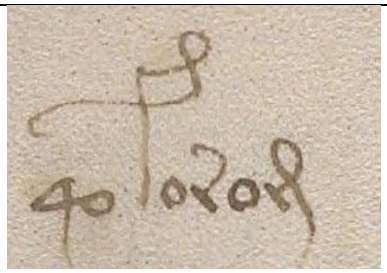
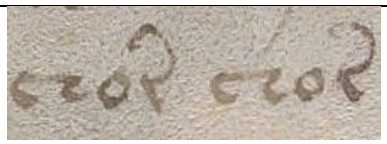
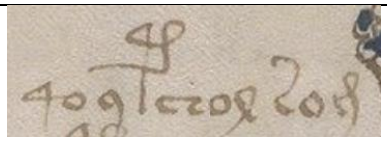
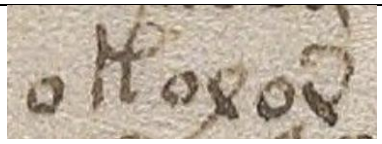
identify further instances which were legible based on the letters identified in POLAR. The Pleiades or Seven Sisters are immediately apparent from the illustration and while we later noticed that Bax [11] had correctly proposed the association of this illustration with the Taurus constellation, we independently came to the same conclusion. However, we read the letter sequence as TOARE not as TAURN [11]. Once it became clear that "o" at the beginning of words is often an article (as is common in some early vernacular romance languages), this allowed for the reading of more words. O'CETE likely refers to the constellation of Cetus and O'COR(vo) possibly refers to the constellation of Corvus (Figure 1d, 1e). Both constellations are featured in the Almagestum and were thus known at the time of the manuscript's creation [23].

Figure 1: Initially identified terms for celestial bodies and constellations. A) A star illustration joined by the letters POLAR indicating the North Star. B) A star illustration with the letters EXPE (possibly shortened for Experia) a name for the Evening Star Venus. C) The Pleiades cluster with the letters TOARE (Taurus) indicating its affiliation with the constellation of the bull. D) A star illustration with the letters O'CETE (possibly the Cetus constellation). E) A star illustration with the letters O'COR (possibly shortened for the Corvus constellation).

Based on these letters we were then able to develop a partial alphabet which features the most frequent characters and ligatures and allows for an Italian vernacular reading of the text. We have not been able to unequivocally decipher some of the rare characters, but the frequently occurring letters allow to read the majority of words. Folio 66r of MS408 features an itemized list, where individual characters stand for each item in the list. The assumption that these are all individual characters, allowed us to identify the letter M which so far had often been read as two or even three separate glyphs.

Table 1: The key to a lingua volgare reading of MS408

Glyph	Meaning	Explanation	Vocabulary Example
a	A	Latin letter “a” like in the word “polar” (Polaris, North Star).	
ꝛ	B	A rare character in the manuscript, often obscured by contractions. Here the word “arbor” (tree).	
ꝯ	C	One of the so-called gallows characters, denoting “c” here in the word “color” (color).	
ꝺ	D	One of the so-called gallows characters denoting “d” here in the word “odore” (smell).	
9	E	The number 9, a mirrored “e” like in the word combination “piente” (abundant).	
ꝓ	F/P	A greek p, pronounced “f” like in the word “o’c(e)fal” (head) or in the word “polar”.	
ꝯ	G	A rare character, often obfuscated by contractions. Here o’g(a)mo (marriage, sex). Identification remains uncertain.	

ꝛ	I	A letter potentially confusable with the letters for “r” and “g”. Here the word combination “p(er)vivi” (for the living)	
ꝶ	L	A lambda standing for “l” like in the word “oliar” (to oil).	
ꝵ	M	A difficult letter to read, like here in “compie” (to complete / to fill). It is often split in two or even three parts and therefore misidentified as several glyphs.	
ꝶ	N	Like “m” a difficult letter to read as the line between two parts is often obscured leading to it being read as “a-r”. Here the word “fornare” (to bake, to roast).	
ꝏ	O	Latin letter “o”. Like in the word “sorore” (sister, nun). “o” often appears in front of nouns as an article or in front of verbs as a pronoun.	
ꝓ	P/F	A greek p, pronounced “p” like in the word “porpor” (purple, crimson).	
4	Q	A “4” almost always followed by the letter “o” forming the word “qo”. Here the word combination “qo exfoliare” (which to exfoliate). Also see “qo sorore” above.	
ꝛ	R	A left-tilted small “r” as in “o’dolor” (pain).	

	S	Like the letter “x” but with only one loop on top, like here in the word “o’sole” (the sun).	
	T	The number 8. Here the word “torpore” (dizziness).	
	U	Written as a “c” like here in o’cute (skin). The identification is particularly clear with the word “qo” which is usually written without a “u” but appears occasionally written as “quo” (e.g. folio 43r line 13)	
	V	Closely written double “c” like in the word “ovui” (eggs).	
	X / SS	Like the letter “s” but with two loops on top. Here the word “exfol(ie)” (without leaves).	
Special Characters			
	M~	The letter “m” followed by an additional stroke indicating the syllable “mo”. Here the word “palmo” (palm). The glyph most frequently occurs as a verb ending indicating 1st person plural present tense.	
	N~	The letter “n” followed by an additional stroke indicating the syllable “no” Here the word “c(a)none” (canon).	

	P/F	A “p” or “f” sound with a shortening sign on top. This glyph exclusively stands for i-related syllables like “pi”, “fi”. Here the word “fiore” (flower).	
Ligatures			
	Re	Ligature at the ending of words meaning “re” a combination of both letters like here in the word “o’colore” (the color).	
	Te	Similar to the re-ligature but with a rounded left side. A combination of the letters “t” and “e”. Here in the word “alte” (high).	
	CP	A ligature consisting of the letter c and the letter p/f like here in the word “c(e)pol” (onion).	
	DP/DF	A ligature consisting of the letter d and the letter p/f like here in the word combination “de folie” (of leaves).	
	SP	A ligature consisting of the letter s and the letter p/f like here in the word “s(e)parare” (to divide).	
	XP	A ligature combining the letter x and the letter p/f like here in the word “expuimo” (to spit out?)	

This simple substitution cipher allows for reading the majority of words in MS408. The missing letter H is not required for vernacular Italian and we have not identified any suitable candidates among the glyphs that could represent this letter. J and K can be represented by I and C respectively. W does not exist in Latin nor in vernacular Italian. In Latin Y is only used for Greek loanwords. Z is often written as T in vernacular texts and therefore not necessarily needed here either.

Based on this reading of the glyphs, we have created a list of 400+ words and word combinations which can be accessed in the *Supplementary Information*. It is important to note that we consider all of these identifications as potential and preliminary. The text is written in shorthand and thus words do need context for their proper identification. While some words are fully rendered, the majority of them are contracted and MS408 features some extreme contractions for the most common words as is the case for other medieval manuscripts written in shorthand.

Discussion

There have been various and ongoing attempts to produce a transliteration of MS408. R. Zandbergen provides an excellent overview over the history of development of the transliterations in [24]. We, however, decided to only work with the manuscript. The reason for this decision is that transliterations like the Extended Voynich Alphabet (EVA) and Voynich101 (v101) go into a great amount of detail which can obscure the meaning and ambiguity of individual glyphs: In both EVA and v101, some glyphs which are a unit, are broken up, some glyphs which should be split, are represented together, and others which are merely variations of the same letter, are featured separately. These problems and the resulting redundancy might also be a reason – in addition to the text being written in shorthand – why computational approaches which rely on machine readable transliterations have so far not been successful.

The vocabulary which we were able to generate, maybe unsurprisingly for a manuscript featuring herbal illustrations, features a significant number of words related to plants. Plant parts like *fiore* (flower),

tal(l)o (shoot), *pomo* (fruit), *ramo* (branch) and *folie* (leaves) are among the most frequent words. The word flower for example occurs in different spellings as *fior*, *fiore*, and *fiuor*. This adds support to the idea that the manuscript was written by multiple authors [cf. 15] possibly from different parts of Italy with diverging dialects. The word for leaf/leaves most commonly occurs in its abbreviated form *fol*, but occasionally is fully rendered as *folie*. We have made a number of potential identifications of plant names such as *pino* (pine), *orno* (mountain ash, *sorbus aucuparia*), *pion(ia)* (peonies), *sofor(a)* (*papilionacea*), and *iss(o)pe* (hyssop, *hyssopus officinalis*). Related adjectives and adverbs like *opimo* (fertile), and *i(e)male* (overwintering) are featured in the text and a number of other terms relating to vegetation and horticulture like *arbor* (tree), *tal(l)ete* (sucker nursery), and *olivete* (olive groves) can be found. We also identify a number of verbs which directly relate to horticulture like *potare* (to trim, to cut), *piotare* (to cover with sods), *arare* (to plough), *colonare* (to reclaim land), and possibly also *t(err)are* (to cover with soil / to plant). While many of these terms are only potential identifications due to the inherent ambiguity of a shorthand text, we believe that the quantity of such terms is a strong argument for the identification of the MS408 language as vernacular Italian.

Although the characters used have to be considered an encrypted alphabet, it is a simple substitution cipher with separate letters and a small number of ligatures. The problem is that frequently used words are written in a contracted form – a common practice in medieval manuscripts. The manuscript uses abbreviations for many words, often indicating only the initial part of a word or the letters forming its root. Nonetheless, we were able to identify some grammatical and morphological features. At this stage we can only preliminarily identify some aspects which give further clues regarding the language identification:

A preceding “o” stands for an article when the word is a noun, consistent with many early Romance languages. This is most clearly illustrated by the word *color(e)* and *o’colore* in Table 1. There are a number of examples where words appear both with and without the article throughout the text. The article “o” precedes both masculine and feminine singular nouns. This practice of attaching the article to the noun is also seen in 15th century herbal manuscripts like the Erbario della biblioteca Gambalunga [25].

The MS408 language furthermore features additional preceding agglutinations like the word *qo* (which/that) which is added to the front of a noun or verb, resulting in many terms starting with the otherwise unusual letter combination *qo*. Another likely prefix is the word *d(e)* (of). There are potentially other agglutinating syllables like *(e)x* (from) and *ol* (to the). Different from modern Italian, MS408 seems to use just one articulated preposition "ol" (instead of al, allo, alla, ai, etc.) and likely one pronoun for feminine and masculine singular nouns.

We have not been able to fully make sense of the verbs in the text. Most forms appear to be written in present tense first person plural like in the cases of *pot(a)mo* (we cut/trim), *alt(a)mo* (we grow/increase), *sort(a)mo* (we extract), *foc(a)mo* (we set on fire), etc. The verb *essere* seems to often be hidden unless it occurs in other modes or tenses (*fui, fusse)*. The spelled-out forms of *essere* indicate that we are closer to vernacular Italian than to Late Latin. An omitted or implied verb "to be" is also common practice in many vernacular Italian texts.

We also identify a number of word accumulations which create meaning together (Figure 2). Given the uncertainty regarding the word identifications, these have to be considered with the appropriate skepticism, but they seem to follow a grammatical structure which is in agreement with an identification of the language as vernacular Italian. It seems unlikely, that a cipher would yield a meaningful word combination like *o' d(e)po(r)vi ovui* (to lay eggs into) if it were to be completely wrong (Figure 2c). Furthermore, these word combinations allowed us to identify similar texts which occasionally even feature the same abbreviations but are not written in cipher. The Carta Balanina (13th century) [26], the Letters of Andrea di Bartolomeo di Ghino (14th century) [27], and Ghino's herbal manuscript (first half 15th century) [28] show for example the same use of *n(on), pche/pce* for *p(er)c(h)e,* and a single agglutinated *p* for *p(er).* They also feature a plethora of words which are contracted the same way as the vocabulary of MS408: *bott(ega), cint(ali), co(m)p(er)ate, p(ar)te,* and others that reveal the same type of shorthand.

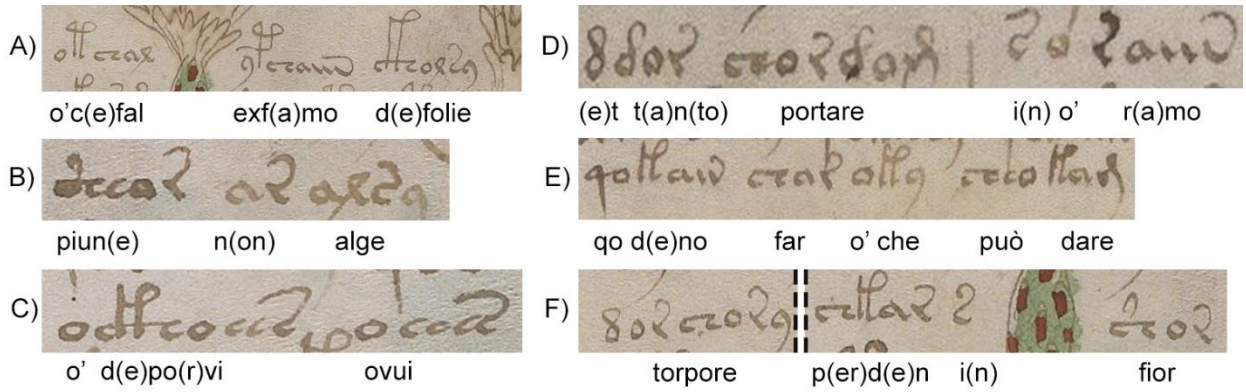


Figure 2: Examples of words which can be read in context: A) folio 2r line 1. B) folio 88r line 13. C) folio 24r line 15. D) folio 102r line 11-12. E) folio 106r line 47. F) folio 2r line 2 (dashed lines exclude the illustration interrupting the text).

Finally, we can hypothesize based on the vocabulary we identified and the historical authors for which we found the best vocabulary matches that MS408 might have originated somewhere in central Italy The main authors we find use similar words as in MS408 are Dante Alighieri (Firenze 1265 – Ravenna 1321), Torquato Tasso (Sorrento 1544 – Roma 1595), Giovanni Boccaccio (Certaldo 1313 – Certaldo 1375), Francesco Petrarca, (Arezzo 1304 – Arquà 1374), Ludovico Ariosto (Reggio Emilia 1474 – Ferrara 1533), and Jacopone da Todi (Todi ca. 1230-1236 – Collazzone 1306) (see *Supplementary Information*). The majority of authors in whose texts we find correspondences to the MS408 vocabulary stem from Tuscan cities like Firenze, Siena, and Pisa. Especially in the 14th and 15th century, a large number of herbal manuscripts was produced in Italy often in volgare or in a Latin full of vernacular terms [29] – the MS408 belongs to this group of manuscripts.

Conclusion

After more than one hundred years of unsuccessful attempts at deciphering, the key we provide here finally allows for an identification of the language of MS408 as an Italian vernacular written in shorthand. Given the frequent claims of deciphering, we want to make it clear that we see the proposed alphabetic cipher as preliminary and potentially needing substantial corrections as more words are read in context. However, we uncovered a rich vocabulary related to plants and horticulture consistent with the illustrations seen on

many pages. MS408 relies on a simple substitution cipher where each glyph stands for a letter of the Latin alphabet with the exception of some ligatures and abbreviated frequently occurring syllables. The use of a shorthand characteristic of medieval manuscripts in combination with the cipher and the agglutinating reductive characteristics of the manuscript is likely the reason why it has proven to be so challenging to read MS408. While ambiguities remain and there certainly are improvements to be made with regards to the identifications of vocabulary, we trust that this article is able to provide the basis for future multidisciplinary studies leading to a proper translation of the Voynich Manuscript.

Methods

Through an initial identification of the unabbreviated words "polar" and "torpore" we were able to associate some MS408 glyphs with characters of the Latin alphabet. This reduced the options space for potential languages consistent with the 15th century date of the manuscript significantly. We hence worked with the assumption that the text is written in a European language and more likely a romance language like Late Latin, vernacular Italian, Catalan, Occitan, or French. We then followed a discursive approach: using text messaging apps the authors discussed the identifications of individual words and expanded the set of characters available. MS408 features an enumeration list on folio 66r which uses individual glyphs. This was helpful in identifying individual characters. The glyphs *for m, n, mo,* and *no* were particularly difficult because they are often split into several parts and not easily recognizable as one unit. Rare letters such as *b* and *g* took longer to identify and some uncertainty remains as only few words feature them. Some ligatures also remain unclear.

We then proceeded to identify individual words throughout the entirety of the manuscript with the aim of finding readable words on every folio. Reading some of the words in context with other words and in context with the illustrations allowed us to revise previous assumptions. We created a list of potential vocabulary (see *Supplementary Information*). As MS408 is written in shorthand, many of the individual identifications remain ambiguous. The context can help reduce this ambiguity, but the words supplied in

the lexical list have to be understood as potential identifications and not as unequivocal readings. We used solely words which could be identified through the *Tesoro della Lingua Italiana delle Origini* [30], the online dictionary of the *Accademia della Crusca* [31], and *Treccani Enciclopedia della Lingua Italiana* [32] to substantiate word identifications. The best identifications are words which have only one or no contraction and allow for an obvious direct correlation with an illustration on the same folio. Ultimately, the full context including grammatical structures will need to be taken into account in order to substantiate the individual identifications. We have provided quotes from authors roughly contemporary with the manuscript for each word identification, which allows for the words to be read in context in a contemporary source. Each of the quotes also features other words occurring in the manuscript, showing that these words form part of a coherent vocabulary.

Shorthand manuscripts are notorious for their ambiguity with regards to word endings and extreme contractions. It is possible to create a grammatically correct text in vernacular Italian, however, some uncertainties remain regarding the identifications of individual words due to the abbreviated nature of many terms.

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Competing Interest

The authors declare no competing interests.